1.a)

$$\left(\begin{array}{ccc|c} -3 & -1 & \alpha & -4 \\ -2 & 0 & 2 & -6 \\ 0 & -3 & -6 & 15 \\ 4 & \alpha & -2 & b \end{array}\right) \begin{matrix} \\ \\ \\ \\ \end{matrix} \begin{matrix} \\ \curvearrowleft -2 \\ \\ \curvearrowleft +2 \end{matrix} \sim \left(\begin{array}{ccc|c} 1 & -1 & \alpha-4 & 8 \\ -2 & 0 & 2 & -6 \\ 0 & -3 & -6 & 15 \\ 0 & \alpha & 2 & b-12 \end{array}\right) \begin{matrix} \curvearrowleft +2 \\ \\ \\ \curvearrowleft \times(-\frac{1}{3}) \end{matrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & -1 & \alpha-4 & 8 \\ 0 & -2 & 2\alpha-6 & 10 \\ 0 & 1 & 2 & -5 \\ 0 & \alpha & 2 & b-12 \end{array}\right) \begin{matrix} \\ \\ \curvearrowleft 2 \\ \curvearrowleft -\alpha \end{matrix} \sim \left(\begin{array}{ccc|c} 1 & -1 & \alpha-4 & 8 \\ 0 & 0 & 2(\alpha-1) & 0 \\ 0 & 1 & 2 & -5 \\ 0 & 0 & 2(1-\alpha) & b-12+5\alpha \end{array}\right) \begin{matrix} \\ \curvearrowleft \\ \curvearrowleft \\ \curvearrowleft +1 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & -1 & \alpha-4 & 8 \\ 0 & 1 & 2 & -5 \\ 0 & 0 & 2(\alpha-1) & 0 \\ 0 & 0 & 0 & b-12+5\alpha \end{array}\right)$$

No solutions: If $b-12+5\alpha \neq 0$.

One solution: If system is consistent and has no free variables: $b-12+5\alpha=0$ and $2(\alpha-1)\neq 0 \ (\Leftrightarrow \alpha \neq 1)$

Infinitely many Solutions: If system is consistent and has free variables: $b-12+5\alpha=0$ and $2(\alpha-1)=0$.

So $\alpha=1$ and $b=7$.

b) Substituting $\alpha=1$ and $b=7$ in the echelon form of (a) gives

$$\left(\begin{array}{ccc|c} 1 & -1 & -3 & 8 \\ 0 & 1 & 2 & -5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \begin{matrix} \\ \\ \\ \curvearrowleft +1 \end{matrix} \sim \left(\begin{array}{ccc|c} 1 & 0 & -1 & 3 \\ 0 & 1 & 2 & -5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\Rightarrow \begin{cases} x_1 - x_3 = 3 \\ x_2 + 2x_3 = -5 \\ x_3 = x_3 \text{ (free)} \end{cases} \quad \text{So} \quad \vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_3 + 3 \\ -2x_3 - 5 \\ x_3 \end{pmatrix} =$$

$$= \begin{pmatrix} 3 \\ -5 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} x_3 = \underbrace{\begin{pmatrix} 3 \\ -5 \\ 0 \end{pmatrix} + \text{Span}\left\{\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}\right\}}_{\text{solution in parametric vector form.}}.$$

c) From the reduced echelon form in (b): Columns 1 and 2 have pivots $\Rightarrow$ these are linearly independent. Thus a basis of $\text{Col}A$ is $\left\{\begin{pmatrix} -3 \\ -2 \\ 0 \\ 4 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ -3 \\ 1 \end{pmatrix}\right\}$ and $\dim \text{Col}A = 2$.

2. a) This volume is $|\det A|$.

$$\det A = \begin{vmatrix} 4 & 6 & 0 \\ 1 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} \overset{\cdot(-4)}{=} \begin{vmatrix} 4 & 6 & 0 \\ -3 & -5 & 0 \\ 1 & 2 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 4 & 6 \\ -3 & -5 \end{vmatrix} =$$

$$= -20 - (-18) = -2. \quad \text{So the volume is } |-2| = 2.$$

b) $\det A \neq 0$ (from (a)) so A is invertible.

We find A^{-1} in the reduced echelon form of $(A \mid I)$

$$\left(\begin{array}{ccc|ccc} 4 & 6 & 0 & 1 & 0 & 0 \\ 1 & 3 & 4 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 0 & 1 \\ 1 & 3 & 4 & 0 & 1 & 0 \\ 4 & 6 & 0 & 1 & 0 & 0 \end{array}\right) \begin{matrix} \cdot(-1) \\ \sim \\ \cdot(-4) \end{matrix}$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 0 & 1 \\ 0 & 1 & 3 & 0 & 1 & -1 \\ 0 & -2 & -4 & 1 & 0 & -4 \end{array}\right) \overset{+2}{\sim} \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 0 & 1 \\ 0 & 1 & 3 & 0 & 1 & -1 \\ 0 & 0 & 2 & 1 & 2 & -6 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & -1/2 & -1 & 4 \\ 0 & 1 & 0 & -3/2 & -2 & 8 \\ 0 & 0 & 1 & 1/2 & 1 & -3 \end{array}\right) \overset{-2}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 5/2 & 3 & -12 \\ 0 & 1 & 0 & -3/2 & -2 & 8 \\ 0 & 0 & 1 & 1/2 & 1 & -3 \end{array}\right)$$

So $A^{-1} = \begin{pmatrix} 5/2 & 3 & -12 \\ -3/2 & -2 & 8 \\ 1/2 & 1 & -3 \end{pmatrix}$.

(You can verify the answer: $A \cdot A^{-1} = I$.)

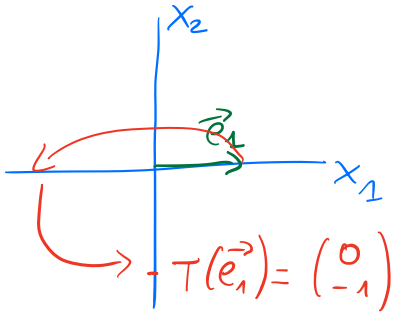
c) Using (b): $\vec{x} = A^{-1} \cdot \vec{b} = \begin{pmatrix} 5/2 & 3 & -12 \\ -3/2 & -2 & 8 \\ 1/2 & 1 & -3 \end{pmatrix} \begin{pmatrix} -2 \\ 5 \\ 1 \end{pmatrix} =$

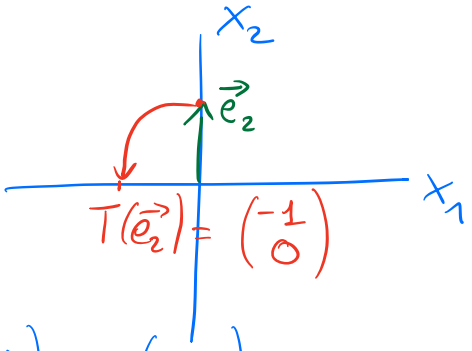
$$= \begin{pmatrix} -5 + 15 - 12 \\ 3 - 10 + 8 \\ -1 + 5 - 3 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}.$$

Note: There is a unique solution because A is invertible.

d) <u>False</u>: In $\alpha \cdot A$ all rows of A are scaled by α. Since A has three rows: $\det(\alpha \cdot A) = \alpha^3 \cdot \det(A)$.

3. a) We apply T to $\vec{e}_1, \vec{e}_2$





$$T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

Thus the representation matrix $A = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

b) $\ker(T) = \{\vec{x} \in \mathbb{R}^2 \mid T(\vec{x}) = \vec{0}\} = \text{Null } A$.

$$\left(\begin{array}{cc|c} 0 & -1 & 0 \\ -1 & 0 & 0 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right) \Rightarrow \vec{x} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus $\ker(T) = \left\{\begin{pmatrix} 0 \\ 0 \end{pmatrix}\right\}$.

c) By (b), $\ker(T) = \{\vec{0}\}$ so T is one-to-one.

$\text{Im}(T) = \text{Col } A = \text{Span}\left\{\begin{pmatrix} -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}\right\} = \mathbb{R}^2$ so T is also onto.

d) Consider an <u>arbitrary</u> vector $\begin{pmatrix} a \\ b \end{pmatrix}$. If $T\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ then $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$. Solve a linear system!

$$\left(\begin{array}{cc|c} 0 & -1 & 3 \\ -1 & 0 & 2 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & -3 \end{array}\right) \Rightarrow \begin{cases} a = -2 \\ b = -3 \end{cases}$$ so

the only vector that has image $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ is $\begin{pmatrix} -2 \\ -3 \end{pmatrix}$.

4. (a) It suffices to show that each of the vectors in $\left\{\begin{pmatrix} 5 \\ -4 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 6 \\ -17 \\ -3 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ 1 \\ 1 \end{pmatrix}\right\}$ belongs to Span S (because then, any linear combination of these three vectors also belongs to Span S). Note that $\begin{pmatrix} 0 \\ 4 \\ 1 \\ 1 \end{pmatrix} \in S$.

Let $A = \begin{pmatrix} 4 & 6 & 0 \\ 1 & 3 & 4 \\ 1 & 2 & 1 \\ -2 & 5 & 1 \end{pmatrix}$. We need to show that the systems $A\vec{x} = \begin{pmatrix} 5 \\ -4 \\ 0 \\ 0 \end{pmatrix}$ and $A\vec{y} = \begin{pmatrix} 6 \\ -17 \\ -3 \\ 0 \end{pmatrix}$ are consistent. For brevety, we use a "joint" augmented coeff. matrix:

$$\left(\begin{array}{ccc|cc} 4 & 6 & 0 & 5 & 6 \\ 1 & 3 & 4 & -4 & -17 \\ 1 & 2 & 1 & 0 & -3 \\ -2 & 5 & 1 & 0 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|cc} 1 & 2 & 1 & 0 & -3 \\ 4 & 6 & 0 & 5 & 6 \\ 1 & 3 & 4 & -4 & -17 \\ -2 & 5 & 1 & 0 & 0 \end{array}\right) \sim$$

$$\left(\begin{array}{ccc|cc} 1 & 2 & 1 & 0 & -3 \\ 0 & -2 & -4 & 5 & 18 \\ 0 & 1 & 3 & -4 & -14 \\ 0 & 9 & 3 & 0 & -6 \end{array}\right) \sim \left(\begin{array}{ccc|cc} 1 & 2 & 1 & 0 & -3 \\ 0 & 1 & 3 & -4 & -14 \\ 0 & 0 & 2 & -3 & -10 \\ 0 & 0 & -24 & 36 & 120 \end{array}\right) \sim$$

$$\left(\begin{array}{ccc|cc} 1 & 2 & 1 & 0 & -3 \\ 0 & 1 & 3 & -4 & -14 \\ 0 & 0 & 2 & -3 & -10 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \leftarrow \text{echelon form(s)}$$

The system is consistent.

Note: other proofs also possible.

b) From (a) we see that in the echelon form each column in A has a pivot.

$\Rightarrow$ The columns of A are linearly independent

$\Rightarrow$ The vectors in S are linearly independent.

c) $[\vec{y}]_S = \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}$ $\rightarrow$ coordinates of basis S

$$\vec{y} = -1\begin{pmatrix} 4 \\ 1 \\ 1 \\ -2 \end{pmatrix} + 2\begin{pmatrix} 6 \\ 3 \\ 2 \\ 5 \end{pmatrix} - 3\begin{pmatrix} 0 \\ 4 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ -7 \\ 0 \\ 9 \end{pmatrix}.$$

5. a) $\vec{x}$ is an eigenvector of A if $A\vec{x} = \lambda\vec{x}$ for some $\lambda \in \mathbb{R}$.

$$A\vec{x} = \begin{pmatrix} 3 & 18 & 0 \\ 1 & 6 & 2 \\ 0 & 0 & 4 \end{pmatrix}\begin{pmatrix} -10 \\ -1 \\ 10 \end{pmatrix} = \begin{pmatrix} -30 - 18 \\ -10 - 6 + 20 \\ 40 \end{pmatrix} = \begin{pmatrix} -48 \\ 4 \\ 40 \end{pmatrix} \neq \lambda\begin{pmatrix} -10 \\ -1 \\ 10 \end{pmatrix}$$

for all $\lambda \in \mathbb{R}$.

So $\vec{x}$ is <u>not</u> an eigenvector of A.

$$A\vec{y} = \begin{pmatrix} 3 & 18 & 0 \\ 1 & 6 & 2 \\ 0 & 0 & 4 \end{pmatrix}\begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 9+18 \\ 3+6 \\ 0 \end{pmatrix} = \begin{pmatrix} 27 \\ 9 \\ 0 \end{pmatrix} = 9\cdot\begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}.$$

So $\vec{y}$ <u>is</u> an eigenvector of A.

b) 4 is an eigenvalue of A if $\det(A-4I)=0$.

$$\det(A-4I) = \begin{vmatrix} -1 & 18 & 0 \\ 1 & 2 & 2 \\ 0 & 0 & 0 \end{vmatrix} = 0.$$ So 4 is an eigenvalue of A.

c) We solve $\det(A-\lambda I)=0$

$$\det(A-\lambda I) = \begin{vmatrix} 3-\lambda & 18 & 0 \\ 1 & 6-\lambda & 2 \\ 0 & 0 & 4-\lambda \end{vmatrix} = (4-\lambda)\begin{vmatrix} 3-\lambda & 18 \\ 1 & 6-\lambda \end{vmatrix}$$

$$= (4-\lambda)\left((3-\lambda)(6-\lambda)-18\right) = (4-\lambda)(18-3\lambda-6\lambda+\lambda^2-18)$$

$$= (4-\lambda)(\lambda^2-9\lambda) = \lambda\cdot(4-\lambda)(\lambda-9).$$

Thus $\det(A-\lambda I)=0$ has solutions $\lambda_1=0$, $\lambda_2=9$ and $\lambda_3=4$.

For the <u>eigenvectors</u> we select a nonzero vector in $Null(A-\lambda_i I)$ for each eigenvalue λ_i:

$\lambda_1=0$ $\quad A-0\cdot I = A = \begin{pmatrix} 3 & 18 & 0 \\ 1 & 6 & 2 \\ 0 & 0 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 6 & 2 \\ 0 & 0 & -6 \\ 0 & 0 & 1 \end{pmatrix} \sim$

$$\begin{pmatrix} 1 & 6 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} x_1+6x_2=0 \\ x_2 = \text{free} \\ x_3 = 0 \end{cases}$$

An eigenvector $\vec{v}_1 = \begin{pmatrix} -6 \\ 1 \\ 0 \end{pmatrix}$.

$\boxed{\lambda_2 = 9}$ $\quad A - 9I = \begin{pmatrix} -6 & 18 & 0 \\ 1 & -3 & 2 \\ 0 & 0 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & -3 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

$\sim \begin{pmatrix} 1 & -3 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} x_1 - 3x_2 = 0 \\ x_2 = \text{free} \\ x_3 = 0 \end{cases}$ An eigenvector $\vec{v}_2 = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$.

$\boxed{\lambda_3 = 4}$ $\quad A - 4I = \begin{pmatrix} -1 & 18 & 0 \\ 1 & 2 & 2 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 2 \\ 0 & 20 & 2 \\ 0 & 0 & 0 \end{pmatrix} \sim$

$\sim \begin{pmatrix} 1 & 2 & 2 \\ 0 & 1 & 1/10 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 18/10 \\ 0 & 1 & 1/10 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} x_1 + \frac{18}{10} x_3 = 0 \\ x_2 = 1/10\, x_3 \\ x_3 = \text{free} \end{cases}$

An eigenvector is $\vec{v}_3 = \begin{pmatrix} -18 \\ -1 \\ 10 \end{pmatrix}$.

d) Take $P = (\vec{v}_1 \ \vec{v}_2 \ \vec{v}_3)$ and $D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$

$D = P^{-1}AP \iff PD = AP$. We show that the latter equality holds:

$PD = \begin{pmatrix} -6 & 3 & 18 \\ 1 & 1 & -1 \\ 0 & 0 & 10 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 27 & -72 \\ 0 & 9 & -4 \\ 0 & 0 & 40 \end{pmatrix}$

$AP = \begin{pmatrix} 3 & 18 & 0 \\ 1 & 6 & 2 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} -6 & 3 & 18 \\ 1 & 1 & -1 \\ 0 & 0 & 10 \end{pmatrix} = \begin{pmatrix} 0 & 27 & -72 \\ 0 & 9 & -4 \\ 0 & 0 & 40 \end{pmatrix}$

$PD = AP$ holds, thus $D = P^{-1}AP$ holds as well.